Panoramic Evaluation of the Effects of Different Isolation Times on Peripheral Blood Mononuclear Cells

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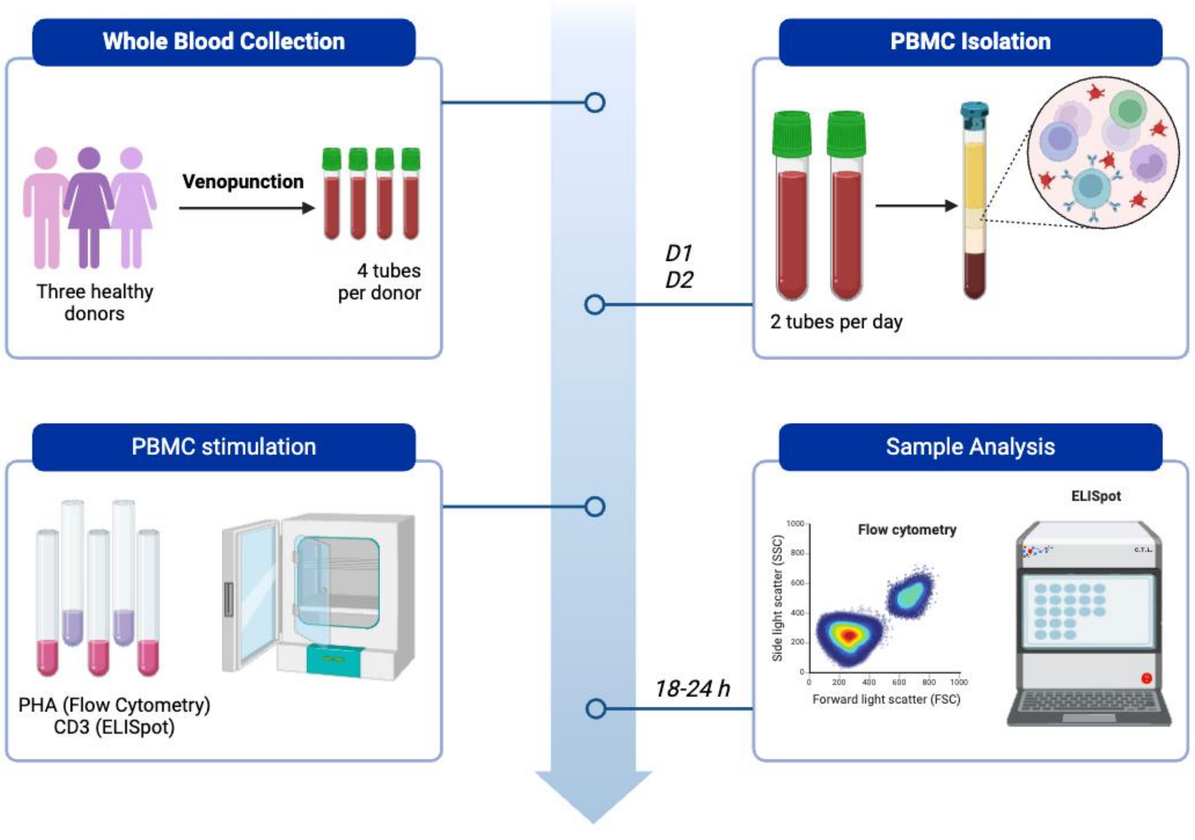
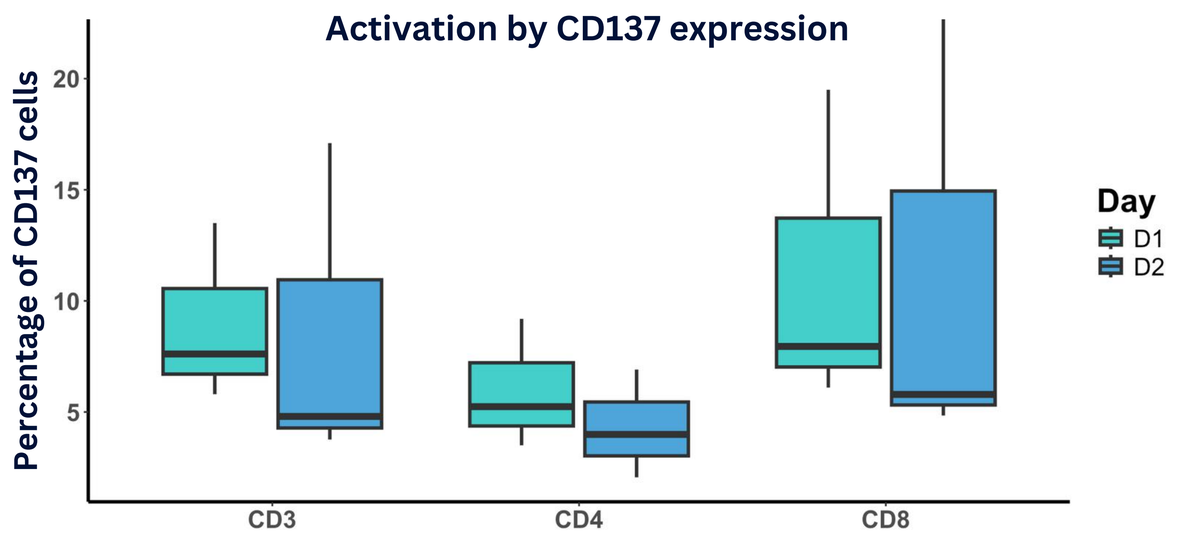
1 INTRODUCTION

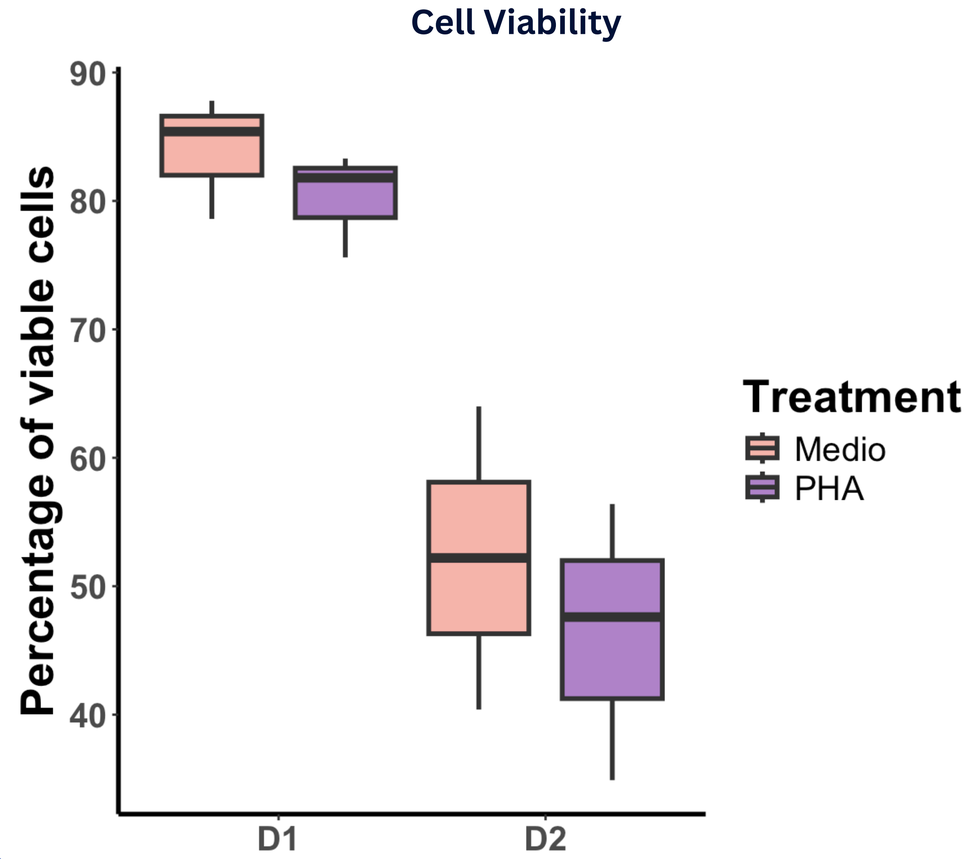
- Human peripheral blood mononuclear cells (PBMCs) are a heterogeneous population of cells which play crucial roles in the immune response and are a valuable resource in research.
- PBMCs are obtained from donors either through whole blood collection or buffy coats and then cryopreserved for long periods.
- They are used in a plethora of research and clinical settings such as evaluation of anticancer drug delivery systems, evaluation of systematic immunity and prediction of responses to immunotherapies, and have emerged as a source of biomarkers in a variety of diseases.
- One major issue in the usage of PBMCs after long term cryopreservation is the reduction in their viability, activation as well as their synthesis and release of cytokines. We evaluated the effect of isolating the PBMCs on different days from their collection while comparing their viability, activation, and release of cytokines after stimulation with CD3 antibody or PHA.

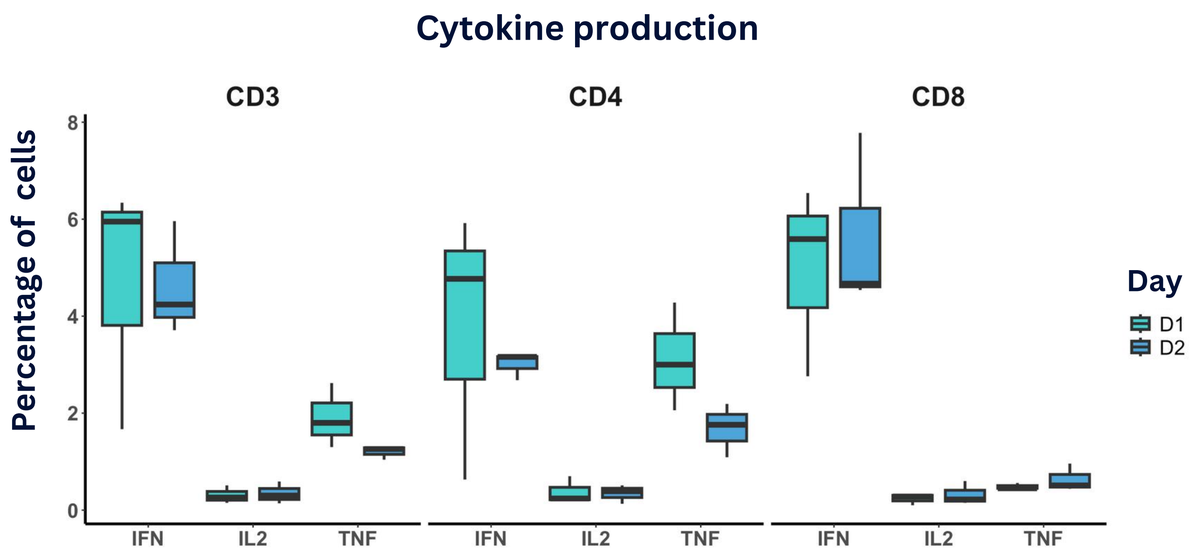
2 METHODS

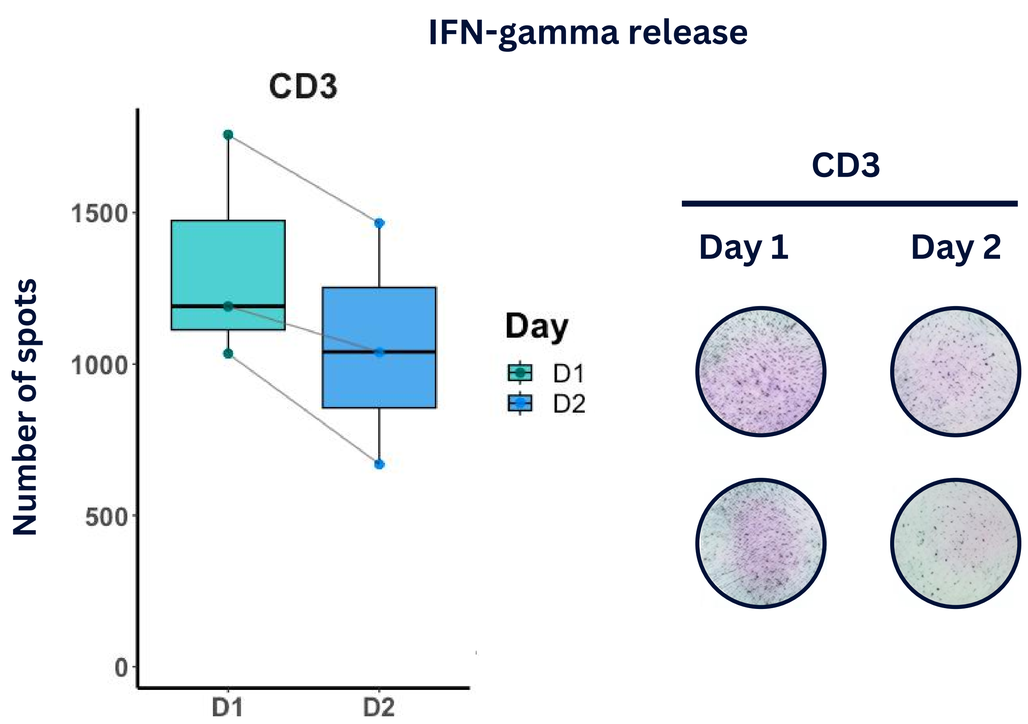


3 RESULTS









4 DISCUSSION

- The activation measured by the expression of CD137 remained unmodified after 24h wait of PBMCs isolation.
- PBMCs exhibited distinct production and release of cytokines when compared from the different days of isolation. In the CD3+ and CD4+ populations there was a tendency of reduced TNF alpha, IFN-gamma, and IL-2 production in PBMCs processed on D2. In the CD8+ subset we found a tendency to increase the production of the studied cytokines on D2.
- The release of IFN-gamma after CD3 stimulation and evaluation by ELISpot showed a decrease.
- In terms of cell viability there was a significant reduction of approximately 40% in D2 when compared with D1 of processing.
- PBMCs must be isolated within the same day of blood sample collection to ensure best results for subsequent experiments as the cell viability is compromised

References